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Towards Trustworthy SBOM Recovery from Compiled Android Applications

An Evidence-Based Component Attribution Framework

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Dedication

[PLACEHOLDER: Personal dedication to be written by the student.]

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Abstract

Software Bills of Materials are increasingly used to support software supply-chain transparency and vulnerability management. However, a generated SBOM is not automatically trustworthy, particularly when it is reconstructed from a compiled Android application without source code or dependency manifests. Compilation, optimization, obfuscation, artifact-family aliases, reference-database gaps, and version similarity can weaken or contradict the evidence supporting a component claim.

This thesis studies trustworthy SBOM recovery as an evidence-based software component attribution problem. It organizes existing Android third-party library and binary component-recovery research into taxonomies of techniques, evidence sources, transformation threats, and evaluation practices. Building on the resulting research gaps, it proposes a conceptual framework in which each recovered claim records component identity, version or version set, evidence provenance, evidence quality, transformation risk, confidence, ambiguity, validation class, and permitted downstream security interpretation.

The implemented APK-to-SBOM PFE system is used as a bounded case study. Its controlled, source-validated, source-constrained, report-only, failed, and limitation evidence demonstrates why library identity, exact version, reference coverage, matcher recall, and vulnerability authority must remain separate. The proposed framework therefore favors auditable partial knowledge and explicit abstention over unsupported certainty, and defines a research agenda for calibrated evidence fusion, provenance-aware SBOMs, realistic benchmarks, and native or dynamically loaded component recovery.

Keywords — software bill of materials, software supply chain, Android, component attribution, provenance, evidence fusion, trustworthiness, uncertainty, vulnerability management.

Résumé

[PLACEHOLDER : La version française sera rédigée après validation de la version anglaise de l'abstract.]

Mots-clés — nomenclature logicielle, chaîne d'approvisionnement logicielle, Android, attribution de composants, provenance, fusion de preuves, confiance, incertitude.

ملخص

سيُستكمل هذا الملخص بعد مراجعة النسخة الإنجليزية واعتماد المصطلحات العلمية المتعلقة بمكونات البرمجيات والأدلة والثقة.

الكلمات المفتاحية — مكونات البرمجيات، سلسلة توريد البرمجيات، أندرويد، الأدلة، مصدر المعلومات، الثقة، عدم اليقين.

Contents

Dedication i

Acknowledgements ii

Abstract iii

Résumé iv

ملخص v

Contents vi

List of Figures x

List of Tables xi

List of Acronyms xii

I General Introduction 1

1 General Introduction 2

1.1 Context . . . 2
1.2 Problem Statement . . . 2
1.3 Motivation . . . 2
1.4 Research Questions . . . 2
1.5 Objectives and Contributions . . . 3
1.6 Scope . . . 3
1.7 Thesis Organization . . . 3

II Background and Research Methodology 4

2 Background and Research Methodology 5

2.1 Introduction . . . 5

2.2 Software Supply Chains and SBOMs 5
2.2.1 Authoritative and Reconstructed SBOMs 5
2.3 Trust, Evidence, Provenance, and Uncertainty 6
2.3.1 Evidence and Claims 6
2.3.2 Provenance and Independence 6
2.3.3 Uncertainty and Abstention 6
2.4 Android Compiled-Artifact Context 6
2.5 Software Component Attribution 7
2.5.1 Identity, Version, and Security Authority 7
2.5.2 Reference Dependence 7
2.6 Evidence Sources in APKs 7
2.7 Structured Literature Review Methodology 8
2.7.1 Review Questions and Search Strategy 8
2.7.2 Eligibility and Screening 8
2.7.3 Evidence Tiers 8
2.8 Analysis Dimensions 9
2.9 Threats to the Review 9
2.9.1 Search and Selection Bias 9
2.9.2 Availability and Recency 9
2.9.3 Heterogeneous Evaluation 9
2.9.4 Reviewer and Extraction Bias 10
2.10 Conclusion 10

III State of the Art and Taxonomy 11

3 State of the Art and Taxonomy 12
3.1 Introduction 12
3.2 Taxonomy of Component-Recovery Techniques 12
3.3 Taxonomy of Evidence Sources 14
3.4 Taxonomy of Transformation Threats 15
3.5 Representative Android TPL Approaches 17
3.5.1 From Profile Matching to Structural Boundaries 17
3.5.2 Version-Aware and Vulnerability-Oriented Pipelines 17
3.5.3 Learning-Based Presence Detection 18
3.5.4 Empirical Assessments and Cross-Tool Studies 18
3.6 Evaluation Practices and Ground Truth 18
3.7 Comparative Analysis 19
3.8 Limitations of Existing Research 21
3.9 Research Gap 22
3.10 Conclusion 23

IV Proposed Framework 24

4 Proposed Evidence-Based Trust Framework 25

4.1 Introduction . . . 25
4.2 Framework Design Principles . . . 25
4.3 Recovered Claim Model . . . 25
4.4 Evidence Hierarchy . . . 26
4.5 Evidence Quality Dimensions . . . 26
4.6 Evidence Fusion, Conflict, and Abstention . . . 26
4.7 Claim Classes . . . 27
4.8 Validation and Evidence Lanes . . . 27
4.9 Vulnerability-Claim Policy . . . 27
4.10 Recovered Claim Lifecycle . . . 27
4.11 PFE System as a Case Study . . . 27
4.12 Framework Evaluation . . . 27
4.13 Limitations and Research Agenda . . . 27
4.14 Conclusion . . . 27

V General Conclusion 28

5 General Conclusion and Future Directions 29

5.1 General Conclusion . . . 29
5.2 Answers to the Research Questions . . . 29
5.3 Main Contributions . . . 29
5.4 Limitations . . . 29
5.5 Future Research . . . 29
5.6 Closing Remarks . . . 29

A Literature Review Protocol 30

A.1 Research Questions . . . 30
A.2 Search Strategy . . . 30
A.2.1 Sources and Scope . . . 30
A.2.2 Concept Blocks . . . 31
A.3 Inclusion and Exclusion Criteria . . . 31
A.3.1 Inclusion . . . 31
A.3.2 Exclusion . . . 32
A.3.3 Screening and Evidence Tiers . . . 32
A.4 Data Extraction Schema . . . 32
A.5 Quality Assessment . . . 33
A.6 Synthesis Rules . . . 33

A.7 Selected Studies 34
A.8 Audit and Change Control 35

Bibliography **36**

List of Figures

4.1 Conceptual flow of the proposed evidence-based trust framework. 26

List of Tables

3.1 Technique taxonomy for Android component recovery. 13
3.2 Evidence-source taxonomy for reconstructed Android component claims. . 14
3.3 Transformation-threat taxonomy for Android component recovery. 16
3.4 Evaluation-practice taxonomy in the Android TPL literature. 19
3.5 Tier A studies classified by recovery role and trust implication. 20

4.1 Initial evidence hierarchy for reconstructed component claims. 26

A.1 Search concept blocks 31
A.2 Structured extraction fields 32
A.3 Frozen seed corpus and current evidence status 34

List of Acronyms

Acronym	Definition
APK	Android Package Kit
API	Application Programming Interface
CVE	Common Vulnerabilities and Exposures
DEX	Dalvik Executable
ML	Machine Learning
NVD	National Vulnerability Database
OSV	Open Source Vulnerabilities
PURL	Package URL
R8	Android code shrinker and optimizer
SBOM	Software Bill of Materials
SCA	Software Composition Analysis
SPDX	Software Package Data Exchange
TPL	Third-Party Library

Part I

General Introduction

Chapter 1

General Introduction

1.1 Context

Software supply chains depend on large ecosystems of reusable components. SBOMs improve transparency by recording these components, but many audit scenarios provide only compiled software artifacts. In such cases, an SBOM is reconstructed from indirect and transformed evidence rather than generated from authoritative dependency metadata.

1.2 Problem Statement

A recovered component name or version is not trustworthy merely because a tool emitted it. Trust depends on the origin, independence, transformation resilience, version specificity, and validation status of its evidence. Existing Android detection research provides strong technical signals, but lacks a unified model connecting those signals to claim provenance, uncertainty, and permitted vulnerability interpretation.

1.3 Motivation

1.4 Research Questions

1. Which evidence sources are available for recovering software components from compiled Android applications?
2. How should evidence strength for identity and exact-version attribution be assessed?
3. How should conflicting, missing, transformed, or source-dependent evidence be represented?
4. What information must accompany a reconstructed SBOM claim for it to be auditable and useful in vulnerability management?

1.5 Objectives and Contributions

The thesis contributes a taxonomy of recovery techniques, an evidence and transformation taxonomy, an evidence-based claim model, a validation-lane model, a vulnerability-authority policy, and a research agenda illustrated through selected evidence from the associated PFE project.

1.6 Scope

1.7 Thesis Organization

Part II

Background and Research Methodology

Chapter 2

Background and Research Methodology

2.1 Introduction

This chapter establishes the concepts and review method used to reason about trustworthy recovery of software components from compiled Android applications. The thesis treats a recovered SBOM as a set of evidence-backed claims rather than as a simple list of names and versions. This distinction requires a common vocabulary for component identity, provenance, uncertainty, and transformation risk, followed by a transparent method for selecting and comparing prior research.

2.2 Software Supply Chains and SBOMs

Modern applications assemble code, resources, native binaries, generated artifacts, and transitive dependencies from many independently maintained projects. An SBOM records these components so that operators can reason about inventory, licensing, provenance, and known vulnerabilities. Standards define how such information is represented, but a representation format does not establish that the recorded component is actually present or that its exact version is known.

The framework considers both SPDX and CycloneDX, while using Package URL as a standard package-identity model [1, 2, 3]. These standards therefore define the target vocabulary and exchange structures, while the attribution framework developed in this thesis defines the evidential conditions under which individual fields may be populated.

2.2.1 Authoritative and Reconstructed SBOMs

An authoritative build-time SBOM can use dependency resolution metadata, lockfiles, build graphs, and package-manager records. By contrast, an artifact-reconstructed SBOM begins after those authoritative records have been removed or made unavailable. It must infer a component from observable residue in an APK, compare that residue with references, and preserve the uncertainty introduced by compilation and incomplete coverage.

The distinction is central to this thesis. A reconstructed component identity may

be credible even when its exact version is unresolved. Similarly, a syntactically valid CycloneDX or SPDX document may contain semantically weak claims. Trustworthiness is consequently a property of each claim and its evidence trail, not of the file format alone.

2.3 Trust, Evidence, Provenance, and Uncertainty

2.3.1 Evidence and Claims

An evidence item is an observable fact used to support or challenge a component claim. Examples include a package fragment, an opcode fingerprint, a class-dependency relation, a resource marker, a native-library name, or a source-validated expected dependency. A claim is the interpretation built from one or more such items, for example that a library family is present or that a particular version is the best-supported candidate.

2.3.2 Provenance and Independence

Provenance records where evidence came from, how it was extracted, and which transformations may have affected it. Two matching fields are not independent when they are derived from the same underlying artifact feature. Confidence must therefore not be increased merely by counting correlated signals as separate confirmations.

2.3.3 Uncertainty and Abstention

Uncertainty can arise from missing evidence, conflicting evidence, incomplete reference coverage, adjacent versions with near-identical code, partial library inclusion, or destructive optimization. The framework allows ambiguity sets and abstention because an explicit unresolved result is more auditable than a forced exact version. This policy also limits downstream vulnerability authority: a vulnerability finding cannot be stronger than the identity and version evidence on which it depends.

2.4 Android Compiled-Artifact Context

Android compilation and R8 transformations directly affect the stability and interpretability of recovered evidence [4].

Java or Kotlin source is compiled and transformed into DEX bytecode, while resources and manifests are packaged into an APK. Dependency boundaries that were explicit during the build may disappear when classes are merged into one or more DEX files. R8 can rename identifiers, remove unused code, rewrite control flow, merge classes, inline methods, and change call sites. Packaging may also introduce multidex layouts, shaded or relocated code, native libraries, generated classes, and dynamically loaded components.

These transformations do not affect all evidence equally. Names and package trees are inexpensive but fragile. Opcode, control-flow, call, and dependency structures can survive renaming, yet they are still altered by shrinking and optimization. Resources and metadata can provide independent support, but they may identify only a family rather

than an exact version. The state-of-the-art taxonomy in chapter 3 therefore classifies both evidence sources and the transformations that threaten them.

2.5 Software Component Attribution

2.5.1 Identity, Version, and Security Authority

Component attribution contains at least three distinct decisions:

1. **identity attribution**: whether a component family or coordinate is present;
2. **version attribution**: which exact version or bounded version set is supported; and
3. **security interpretation**: which vulnerability statements are permitted by the preceding evidence.

A method that performs well at library-family detection does not automatically provide exact-version authority. Likewise, a vulnerability database match is not confirmation that the affected version is present. Keeping these decisions separate prevents a plausible identity match from being silently promoted into an exact and security-critical statement.

2.5.2 Reference Dependence

Many attribution methods compare an application with profiles extracted from known library releases. Such a method can recognize only what its reference corpus represents. Failure may therefore result from the matcher, the extractor, or a missing reference rather than from absence of the component. Reference coverage is treated as a first-class validity condition throughout the thesis.

2.6 Evidence Sources in APKs

The review extracts evidence into recurring families:

- lexical and namespace evidence, including package and class names;
- bytecode and semantic evidence, including opcodes, API references, literals, and method descriptors;
- structural evidence, including control-flow, call, inheritance, and class-dependency graphs;
- resource and packaging evidence, including manifests, resource names, assets, and native-library files;
- external reference and metadata evidence, including library profiles, Maven coordinates, dependency relations, and release chronology;
- learned representations derived from bytecode or structural features;

- validation evidence, including controlled builds, source manifests, and source-constrained observations.

No family is assumed to be universally authoritative. The framework evaluates its origin, transformation resilience, version specificity, independence, coverage, and validation status.

2.7 Structured Literature Review Methodology

The literature study follows a frozen structured mapping protocol adapted from software-engineering systematic-review guidance [5]. It is not presented as a completed exhaustive systematic literature review because the initial corpus was assembled iteratively before the protocol freeze. Instead, the protocol makes the existing corpus auditable and governs all future search refreshes, screening, extraction, and synthesis.

2.7.1 Review Questions and Search Strategy

The review asks which evidence sources and attribution techniques are used, which Android transformations threaten them, how identity and version results are evaluated, how uncertainty and provenance are represented, and which gaps prevent trustworthy SBOM claims. Searches combine four concept blocks: Android artifacts; third-party components or SBOMs; detection or attribution tasks; and obfuscation, optimization, version, or vulnerability terms.

Primary sources include ACM Digital Library, IEEE Xplore, SpringerLink, ScienceDirect, USENIX, and arXiv. Google Scholar is restricted to discovery and citation chaining. The date scope begins in 2015 and continues through the recorded search date. Full query text and database adaptations are preserved in chapter A.

2.7.2 Eligibility and Screening

Eligible studies propose or evaluate Android binary component attribution, version identification, relevant benchmarks, empirical assessments, or SBOM-oriented recovery frameworks. Studies based only on authoritative source manifests, unrelated malware classification, product pages, duplicates, and records without extractable methodology are excluded.

Screening proceeds through deduplication, title-and-abstract screening, full-text assessment, and evidence-tier assignment. Related conference and journal publications are treated as one study family unless the later publication contributes distinct methods, datasets, or results.

2.7.3 Evidence Tiers

The frozen seed corpus contains 20 publication units. Thirteen have locally available full texts and form *Tier A*. Seven are *Tier B* supplementary records whose metadata or structured notes are available but whose local full texts have not yet been verified. Tier B records may inform search coverage and provisional gap mapping, but they cannot

independently support precise quantitative conclusions. Standards and official platform documentation form contextual *Tier C* and are not counted as primary studies.

2.8 Analysis Dimensions

Each included study is extracted across common dimensions: task granularity, input assumptions, segmentation, evidence families, matching or learning method, transformation coverage, ground truth, dataset composition, metrics, confidence and ambiguity behavior, reference dependence, artifact availability, SBOM output, vulnerability mapping, and stated limitations.

Tier A studies are additionally assessed against eight reporting-quality questions covering objective clarity, reproducibility, ground truth, metric definition, transformation conditions, identity/version separation, artifact availability, and threats to validity. Quality affects the weight of a study in synthesis but does not automatically erase historically important work.

Synthesis is narrative and taxonomy-based. Metrics are compared only when their task unit, dataset, ground truth, and transformation conditions are compatible. Original-paper results remain separate from later independent benchmarks, and library-level, class-level, and version-level results are never merged into a single ranking.

2.9 Threats to the Review

2.9.1 Search and Selection Bias

The pre-protocol corpus was built iteratively and may overrepresent well-known Android security venues or studies connected through citation chains. The frozen search blocks, multiple publisher databases, and explicit refresh log reduce but do not eliminate this risk.

2.9.2 Availability and Recency

Several recent or emerging studies currently exist only as supplementary records. Treating them as Tier B prevents incomplete metadata from acquiring the same authority as inspected full text. Their classifications may change after verification, and all such changes must be logged.

2.9.3 Heterogeneous Evaluation

Reported accuracy values often refer to different task definitions, datasets, library versions, obfuscators, and ground-truth procedures. Direct numerical ranking can therefore be misleading. The review preserves these conditions and prefers qualified comparison over pooled scores.

2.9.4 Reviewer and Extraction Bias

The current review is primarily maintained by one researcher. A fixed extraction schema, explicit `NR` values for unreported information, source citations, and retained structured records provide an audit trail. A later independent screening sample would further strengthen reliability.

2.10 Conclusion

This chapter separated a valid SBOM representation from the evidential strength of its contents and defined component attribution as a sequence of identity, version, and security-authority decisions. It also froze a structured mapping protocol that prevents unavailable full texts, incompatible metrics, and correlated signals from being silently promoted into strong claims. The next chapter applies these rules to build the state-of-the-art taxonomies that motivate the proposed trust framework.

Part III

State of the Art and Taxonomy

Chapter 3

State of the Art and Taxonomy

3.1 Introduction

This chapter applies the structured mapping protocol defined in chapter 2 and chapter A to the Android third-party library detection literature. Its purpose is not to produce a numerical leaderboard. The reviewed studies use different datasets, task units, obfuscation settings, and ground-truth procedures, so a direct ranking would hide the very uncertainty that this thesis aims to expose. Instead, the chapter organizes prior work by technique, evidence source, transformation threat, and evaluation practice.

The analysis is centered on the 13 Tier A studies whose full texts are available locally. Tier B records are useful for search coverage and for identifying emerging directions, especially SBOM-oriented recovery and recent benchmarking, but they are not used here as independent authority for precise quantitative claims. This boundary keeps the review consistent with the protocol and prevents unverified metadata from receiving the same weight as inspected studies.

The chapter uses the term *component recovery* broadly. A study may recover a library family, a package cluster, a class set, an exact version, a vulnerability-related version list, or only a presence signal. For trustworthy SBOM reconstruction, these outputs are not equivalent. A method that detects the presence of a library family does not necessarily justify an exact Maven version, and an exact-version candidate does not automatically justify a CVE claim unless the evidence, reference coverage, and validation context are visible.

3.2 Taxonomy of Component-Recovery Techniques

The reviewed approaches can be grouped into seven recurring technique families. The categories are not mutually exclusive: most mature tools combine several families, for example by using a fast coarse filter followed by a slower structural match. The taxonomy therefore describes the role each family plays in a recovered component claim.

Table 3.1: Technique taxonomy for Android component recovery.

Technique family	Typical evidence	Trust role	Tier A examples
Namespace and metadata heuristics	Package trees, class names, library naming conventions	Cheap candidate generation, but weak authority when names are renamed, flattened, relocated, or shaded.	Assessed in [6, 7]
Profile, hash, and signature matching	Method descriptors, Merkle trees, fuzzy hashes, class or method signatures	Can support identity and, when references are complete, version claims. Its authority is bounded by reference coverage and by the stability of the hashed features under optimization.	LibScout [8], ORLIS [9], LibID [10]
Structural graph segmentation	Call, inheritance, containment, and class-dependency relations	Improves module boundary recovery when package structure is unreliable. Supports stronger identity evidence, but does not by itself solve exact version attribution.	LibD [11], LibPecker [12], assessment studies [6, 7]
Coarse-to-fine matching	Fast filtering followed by detailed opcode, CFG, or structural comparison	Balances scalability and precision. Trust depends on both stages: a missed coarse candidate can cap recall, while an over-permissive fine stage can inflate false positives.	ORLIS [9], ATVHUNTER [13], LibScan [14]
Version-discriminating semantic features	Opcode sequences, basic-block features, call chains, field-operation patterns, literals, and class summaries	Provides the main route from library identity to exact-version authority. Adjacent versions and optimized builds remain difficult because many versions share most code.	ATVHUNTER [13], LibScan [14], LibHunter [15], SAD [16]
Learned representations	Smali token sequences, structural vocabularies, neural embeddings	Can learn obfuscation-resilient presence signals, but the reviewed Tier A learning approach does not provide exact-version or boundary authority on its own.	LibAttention [17]

Technique family	Typical evidence	Trust role	Tier A examples
Empirical assessment and benchmarking	Shared datasets, patched tools, source-known applications, evaluation criteria	Does not directly recover components, but defines which claims are comparable and exposes tool failure modes, missing denominators, and reproducibility limits.	Tool assessments [6, 7], Ramirez thesis [18]

The movement across these families is historically important. Early methods made library profiles and package structure usable at scale, but later studies increasingly treat package names as unreliable evidence. Graph-based and version-aware methods improve the situation by using relations, opcode content, and class behavior. The newest verified Tier A work adds learned representations and finer class-level summaries. Even so, the taxonomy shows that standard SBOM emission remains outside the main output model of the verified Android TPL literature: most studies output names, clusters, versions, class mappings, or vulnerability-linked lists rather than a provenance-rich CycloneDX or SPDX document.

3.3 Taxonomy of Evidence Sources

An evidence source is useful for trustworthy SBOM recovery only when its origin, stability, version specificity, and provenance are understood. The same signal may be strong for identity and weak for exact version. Conversely, a highly version-specific literal may be too fragile to survive resource removal or string encryption. Table 3.2 summarizes the recurring evidence families extracted from the Tier A studies.

Table 3.2: Evidence-source taxonomy for reconstructed Android component claims.

Evidence family	Examples	Typical stability	Trust implication
Names and namespaces	Package prefixes, class names, manifest names	Weak under renaming, package flattening, relocation, and shading	Useful for fast hints and human inspection, but insufficient for exact SBOM claims without corroboration.
Descriptors and API surfaces	Method signatures, fuzzy type descriptors, Android API usage	More stable than names under identifier renaming; weaker under dead-code removal and inlining	Supports identity and coarse version discrimination when combined with reference profiles.

Evidence family	Examples	Typical stability	Trust implication
Opcode and instruction sequences	Basic-block opcodes, method opcode chains, call-chain opcodes	Resilient to naming changes; threatened by optimization, inlining, and control-flow rewriting	Provides version-discriminating evidence, especially when compared against known library releases.
Control-flow and dependency structures	CFGs, call graphs, inheritance graphs, class-dependency graphs	Stable against simple renaming and often against package restructuring; affected by optimization and partial inclusion	Improves module boundary recovery and reduces dependence on package layout.
Literals, constants, and resources	Strings, constants, resource identifiers, assets	Potentially version-specific, but fragile under stripping, string encryption, and resource rewriting	Best used as tie-breakers or independent corroboration, not as sole authority.
Reference corpus and release metadata	Library profiles, Maven coordinates, release chronology, known vulnerable version ranges	External to the APK; stable only if the corpus is complete and correctly constructed	Defines the ceiling for recognizable identities and versions. Missing references must not be interpreted as component absence.
Learned representations	Transformer embeddings over Smali tokens, structural vocabularies	Can absorb complex obfuscation patterns, but may be hard to audit feature by feature	Promising for presence screening and robustness, but provenance and exact version authority require additional explanation.
Validation evidence	Controlled builds, source manifests, F-Droid dependency data, source-known ground truth	Strong within the validation context; not automatically generalizable	Can upgrade a claim's evaluation status, but must remain separate from APK-only inference.

The trust implication of this taxonomy is that evidence fusion should prefer independent and complementary evidence over repeated variants of the same feature. For example, several hash matches derived from the same method body are less independent than a method-level match corroborated by a class-dependency relation and a version-specific literal. The framework in chapter 4 builds on this distinction by treating independence, provenance, and reference coverage as first-class quality dimensions.

3.4 Taxonomy of Transformation Threats

Android component recovery is difficult because compilation and packaging can remove the very boundaries that build-time dependency tools rely on. R8 and related obfuscators

may rename identifiers, flatten packages, remove unused code, rewrite control flow, and inline methods [4]. The reviewed literature shows that no single evidence family remains stable under all transformations.

Table 3.3: Transformation-threat taxonomy for Android component recovery.

Threat	**Evidence most affected**	**More resilient evidence**	**Trust consequence**
Identifier renaming	Class, method, field, and package names	Fuzzy descriptors, opcode features, structural relations	Name matches cannot be treated as independent authority unless corroborated.
Package flattening and relocation	Package-subtree segmentation and namespace heuristics	Class-dependency graphs, call relations, profile matching independent of package roots	Package-based module boundaries become unreliable; graph segmentation becomes more important.
Control-flow rewriting	CFG shape, branch structure, fuzzy CFG hashes	Higher-level dependency relations, method descriptors, class summaries	CFG-only methods may overstate robustness when control-flow transformations are not in the evaluation.
Dead-code removal and shrinking	Presence of optional classes, methods, API calls, literals	Partial-match scoring, class-level confidence, source-known validation	Absence of a feature may indicate transformation rather than absence of the library.
Method inlining and outlining	Method boundaries, call-chain evidence, version-specific method bodies	Multi-scale evidence, class-level summaries, literals, residual dependency structure	Exact-version authority must be capped when inlining changes the features used for version discrimination.
Multidex and split packaging	Single-file DEX assumptions and incomplete scanners	Whole-APK extraction over all DEX files and split artifacts	Evaluation denominators must state whether all executable artifacts were analyzed.
Partial library inclusion	Binary present/absent decisions and complete-profile matching	Similarity scores, per-class evidence, explicit ambiguity classes	A partial match should not automatically become a full component/version claim.
Nested or transitive dependencies	Flat library lists and one-library-one-package assumptions	Dependency graphs, source-assisted validation, relationship-aware SBOM models	A recovered component may be direct, transitive, bundled, shaded, or copied; the relationship must remain uncertain unless evidenced.
Native and dynamic code	DEX-only static features	Native file hashes, load-site evidence, dynamic analysis, source validation	Out-of-scope code must be represented as a boundary, not silently ignored.

The most consequential threats for exact-version claims are shrinking, partial inclusion, and method inlining. Identifier renaming is comparatively well handled by modern structural or opcode-based approaches, but version evidence often depends on small differences between adjacent releases. Optimizations that delete, merge, or inline exactly those differences can turn an apparently confident match into a version-band ambiguity.

3.5 Representative Android TPL Approaches

3.5.1 From Profile Matching to Structural Boundaries

LibScout established a reference-profile model for matching compiled Android applications against known library releases. Its three-layer profile structure and fuzzy method descriptors made identifier renaming less damaging, and its similarity scores already anticipated the need to represent partial evidence [8]. From the viewpoint of trustworthy SBOM recovery, the important lesson is dual: reference profiles can support version claims, but those claims inherit the assumptions of the profile structure and the reference corpus. When package layout or method boundaries are transformed, the confidence of the resulting claim must be reduced rather than silently preserved.

LibD moved away from pure package-layout dependence by recovering library instances through a homogeny graph over inclusion, inheritance, and call relations, combined with opcode-derived fingerprints [11]. This made large-scale library discovery more resilient to naming changes, but the output remained a clustering-oriented library identification result rather than an exact-version or SBOM result. For the taxonomy, LibD is therefore strong evidence for graph-based boundary recovery and weak evidence for version authority.

ORLIS introduced a two-stage obfuscation-resilient design: a coarse filter narrows candidates, and a fine CFG-based fuzzy comparison performs more expensive matching [9]. The design is valuable because it separates scalability from detailed confirmation, and its Jaccard-style similarity naturally supports partial evidence. Its limitations are equally important for the proposed framework: a structural match is not automatically a version claim, and computational cost affects whether a method can be used as routine SBOM evidence.

LibPecker and LibID further illustrate the shift from names to structural and class-level evidence. LibPecker emphasizes class dependency and adaptive similarity, while LibID focuses on reliable identification of obfuscated third-party Android libraries [12, 10]. Together with the later assessment studies, they support the review's central classification: package structure is a fragile feature family, while class relationships, signatures, and opcode-derived evidence are more suitable as identity support under obfuscation.

3.5.2 Version-Aware and Vulnerability-Oriented Pipelines

ATVHUNTER combines class-dependency based decoupling, coarse CFG filtering, fine-grained opcode matching, and a vulnerability database to identify vulnerable library versions [13]. It is one of the most important verified studies for this thesis because it connects version identification to security interpretation. However, its output remains a version and vulnerability list rather than a standard SBOM with explicit provenance, uncertainty, and claim boundaries. The study therefore motivates the framework's rule

that vulnerability authority cannot exceed the strength of the underlying version evidence.

LibScan and LibHunter extend the version-aware family by using richer opcode, class-correspondence, call-chain, and optimization-aware evidence [14, 15]. Their relevance is not only that they improve matching. They also show that code optimization itself is an evaluation dimension. A method evaluated only on clear or lightly obfuscated applications has not demonstrated the same authority as a method evaluated under inlining and shrinking.

SAD provides the most detailed Tier A example of class-structural version-level reasoning. It splits classes into stateful and stateless categories, applies field-operation sequences or opcode similarity accordingly, and uses literals to break ties between adjacent versions [16]. This design supports the thesis claim that exact-version attribution requires evidence at a finer granularity than library presence. At the same time, its remaining sensitivity to method inlining confirms that strong reported version-level performance does not eliminate transformation risk.

3.5.3 Learning-Based Presence Detection

LibAttention represents a different family: instead of comparing explicit profiles, it learns representations from large Smali corpora and predicts library presence with a Transformer-based model [17]. This is important because learned features can capture patterns that rule-based signatures miss under aggressive obfuscation. For SBOM reconstruction, however, the reviewed Tier A result is bounded: it predicts presence rather than exact version, does not recover library boundaries, and does not provide standard SBOM output. In the taxonomy, learned presence detection is therefore a potentially strong candidate-generation signal but not sufficient version or vulnerability authority by itself.

3.5.4 Empirical Assessments and Cross-Tool Studies

The ASE 2020 assessment and the later systematic assessment are not merely background surveys; they define several trust requirements for any recovered SBOM pipeline [6, 7]. They show that tool results depend on module decoupling, multidex handling, AAR processing, obfuscation setting, version-identification capability, efficiency, and ease of reproduction. These studies are a warning against reporting a single accuracy value without denominators, exclusions, and task levels.

The Ramirez thesis adds a source-validation perspective by comparing Android and iOS detection tools and using open-source Android applications as a ground-truth basis [18]. For this master thesis, its main contribution is methodological: source-known applications can validate specific outputs, but that validation is a distinct evidence lane and should not be merged with APK-only inference.

3.6 Evaluation Practices and Ground Truth

The reviewed studies evaluate different questions even when they use similar words such as detection, identification, precision, recall, or version accuracy. A trustworthy synthesis must therefore classify evaluation practice before interpreting the metric.

Table 3.4: Evaluation-practice taxonomy in the Android TPL literature.

Evaluation practice	What it can show	Boundary for SBOM trust
Controlled library or reference corpus	Whether a matcher can distinguish known library profiles or versions under specified transformations	Results depend on reference completeness and may overstate performance on diverse real applications.
Synthetic or tool-based obfuscation tests	Whether selected transformations damage a feature family	Only covers the transformations and configurations tested; R8 inlining and shrinking require explicit treatment.
Market-scale discovery	Whether a detector scales and finds plausible library families in many apps	Without source ground truth, discovery volume is not the same as correctness or exact-version authority.
Open-source or source-known validation	Whether recovered outputs match dependency evidence available outside the APK	Strong for audited cases, but it is a validation lane, not proof that the same APK-only method is generally correct.
Cross-tool benchmark or assessment	Which tools fail under shared criteria and where methodology is inconsistent	Useful for gap analysis, but metrics remain bounded by the benchmark's dataset and task definitions.
Security or vulnerability mapping	Whether detected versions can be connected to vulnerability data	The vulnerability conclusion is valid only as far as the component identity, exact version, and vulnerability database mapping are valid.

This taxonomy explains why the chapter does not pool reported performance numbers. LibScout's profile precision, ORLIS's obfuscation precision, LibD's market-scale discovery volume, ATVHUNTER's version-level results, LibAttention's R8-resilient presence detection, and SAD's class-structural version scores answer different questions [8, 11, 9, 13, 17, 16]. For a reconstructed SBOM, the evaluation question must be restated as: *which claim type is being validated, against which denominator, and under which transformations?*

3.7 Comparative Analysis

Table 3.5 applies the chapter taxonomy to the Tier A seed corpus. The final column states the trust interpretation rather than a simple score. This is deliberate: the same study can be strong evidence for one claim type and weak evidence for another.

Table 3.5: Tier A studies classified by recovery role and trust implication.

Study	Primary role	Main evidence and technique	SBOM-trust implication
LibScout [8]	Reference-profile identity and version matching	Fuzzy method descriptors, package-class-method profile hierarchy, similarity score	Supports the idea of version profiles, but package and method-structure assumptions must be exposed as evidence limits.
LibD [11]	Scalable library discovery and boundary recovery	Homogeny graph, call/inheritance/inclusion relations, opcode-derived fingerprints	Strengthens identity evidence under name obfuscation, but does not provide exact-version or SBOM authority.
LibPecker [12]	Class-dependency based detection	Class dependency hashing, adaptive similarity, weighted class matching	Shows that structural class relations are useful, while similarity still needs explicit thresholds and uncertainty handling.
ORLIS [9]	Obfuscation-resilient structural matching	Coarse filtering, CFG fuzzy hashing, Jaccard similarity	Provides partial-match evidence, but high cost and missing version authority limit direct SBOM use.
LibID [10]	Obfuscated library identification	Signature and structural evidence for reliable identification	Adds evidence for identity under obfuscation; exact version and downstream security authority still require separate support.
ASE 2020 assessment [6]	Cross-tool failure analysis	Unified criteria, tool patches, multidex and AAR observations	Defines required evaluation dimensions and warns that engineering blind spots can dominate measured accuracy.
Systematic assessment [7]	Broader comparative methodology	Six evaluation criteria, module-decoupling comparison, tool availability and limitations	Supports taxonomy-based synthesis and shows that CDG-style evidence is preferable to package-only evidence.
ATVHUNTER [13]	Version and vulnerability-oriented detection	CDG decoupling, CFG coarse filter, opcode fuzzy matching, vulnerability DB	Strongly connects version evidence to security use, but still lacks standard SBOM provenance and full R8-inlining coverage.

Study	Primary role	Main evidence and technique	SBOM-trust implication
LibScan [14]	Precise TPL identification	Class correspondence, method opcode similarity, call-chain evidence	Improves fine-grained matching evidence; claim strength remains tied to reference coverage and transformation conditions.
Ramirez thesis [18]	Cross-platform and source-known evaluation	Tool comparison, F-Droid ground truth, Android/iOS contrast	Shows the value of source validation as a separate lane and confirms that SBOM output is not solved by existing detectors.
LibHunter [15]	Optimization-aware detection	R8/code-optimization analysis, inlining-aware matching ideas	Makes optimization a first-class validity condition for exact-version claims.
LibAttention [17]	Learning-based presence detection	Transformer over Smali tokens and SDK structural vocabulary	Promising for obfuscation-resilient identity hints, but insufficient alone for exact versions, provenance, or CVE authority.
SAD [16]	Class-structural version-level detection	CDG neighbor aggregation, stateful/stateless split, field-operation sequence, literal tie-breakers	Provides strong version-discriminating evidence, while remaining sensitive to method inlining and lacking standard SBOM output.

Three patterns emerge from this comparison. First, the strongest exact-version approaches are multi-evidence approaches: they do not rely on names alone but combine structural, opcode, class, and literal information. Second, evaluation authority is not the same as detection authority. Assessment studies and source-known validation do not identify libraries by themselves, but they determine whether a result can be interpreted safely. Third, standard SBOM trust requirements are largely external to the detector output: provenance, ambiguity, validation class, relationship type, and vulnerability authority are rarely represented as first-class fields.

3.8 Limitations of Existing Research

The taxonomy exposes limitations that recur across the verified literature:

1. **Library identity and exact version are often conflated.** Presence detection, class-set matching, and exact-version attribution are different claims, but tool outputs do not always preserve the distinction.

2. **Reference coverage is rarely treated as part of the result.** A missed match can result from absent references, extraction failure, or a weak matcher. Without coverage metadata, the user cannot tell which failure occurred.

3. **Transformation coverage is incomplete.** Identifier renaming is widely considered, but shrinking, partial inclusion, package flattening, and method inlining remain unevenly evaluated [15, 17, 16].

4. **Confidence is usually a threshold outcome, not an auditable explanation.** Many methods compute similarities or classifier scores, but few expose why a claim should be trusted, which signals conflicted, or why the method did not abstain.

5. **Partial libraries and dependency relationships are not fully represented.** Similarity scores can detect partial inclusion, but a standard SBOM also needs relationship semantics: direct, transitive, bundled, shaded, copied, or uncertain.

6. **Vulnerability interpretation is stronger than the underlying evidence in many pipelines.** A detected library name is not enough for a version-specific CVE claim. Even a plausible version candidate requires reference, transformation, and validation context.

7. **Reproducibility remains uneven.** Tool availability, installation difficulty, dataset access, and benchmark construction vary across studies, limiting independent confirmation of reported results.

8. **Standard SBOM output is not the dominant research artifact.** The verified studies mostly produce detector outputs rather than complete CycloneDX or SPDX statements with evidence, provenance, and uncertainty fields.

These limitations do not make the existing work weak. Rather, they show that the literature has mostly optimized detection performance, while trustworthy SBOM recovery requires an additional layer that explains what each recovered claim means and how it may be used.

3.9 Research Gap

The state of the art provides valuable techniques for recognizing Android third-party libraries from compiled artifacts. Reference profiles, fuzzy descriptors, dependency graphs, opcode features, call-chain evidence, stateful/stateless class summaries, learned Smali representations, and source-known validation all contribute useful evidence. However, the field still lacks a unified model that connects these signals to a reconstructed SBOM claim.

The research gap can therefore be stated as follows:

> Existing Android component detectors provide useful identification and version-discrimination techniques, but they do not consistently attach each recovered SBOM claim to its evidence provenance, reference coverage, transformation risk, confidence boundary, ambiguity, validation class, and permitted vulnerability interpretation.

This gap is not only a formatting problem. A syntactically valid SBOM can still be untrustworthy if it hides uncertainty, promotes identity evidence into exact-version authority, or turns an ambiguous version set into a definitive CVE finding. Conversely, a detector that abstains or reports a bounded version set may be more useful for security decision-making than a pipeline that always emits a single version without explaining the evidence.

3.10 Conclusion

This chapter classified Android component-recovery research by technique, evidence source, transformation threat, and evaluation practice. The synthesis shows a clear technical progression from package and profile matching toward graph-based, version-aware, optimization-aware, and learning-based methods. It also shows a persistent trust gap: detector outputs rarely preserve enough provenance, uncertainty, validation context, and security authority to become auditable reconstructed SBOM claims on their own.

The next chapter responds to this gap by proposing an evidence-based trust framework. The framework does not replace the reviewed techniques. Instead, it defines how their outputs should be represented, fused, bounded, and interpreted so that a recovered component entry communicates not only what was detected, but why it should be believed and how safely it may be used.

Part IV

Proposed Framework

Chapter 4

Proposed Evidence-Based Trust Framework

4.1 Introduction

The proposed framework treats every recovered SBOM entry as a claim that must be accompanied by evidence, provenance, uncertainty, and an explicit boundary on how the claim may be used.

4.2 Framework Design Principles

1. Every claim exposes its supporting evidence.
2. Library identity and exact version are separate claims.
3. Missing evidence is not interpreted as component absence.
4. Conflicting evidence remains visible.
5. Confidence is bounded by transformation and reference-coverage risks.
6. Vulnerability authority cannot exceed component and version authority.
7. Source-assisted validation is distinguished from APK-only inference.
8. Report-only assistance is distinguished from authoritative output.

4.3 Recovered Claim Model

A recovered claim contains the analyzed APK, component identity candidate, artifact family, version or version set, evidence list, evidence provenance, coverage context, transformation risk, confidence class, ambiguity, validation class, and permitted downstream interpretation.

4.4 Evidence Hierarchy

Table 4.1: Initial evidence hierarchy for reconstructed component claims.

Level	Meaning	Permitted interpretation
E0	No component evidence	No package-level claim
E1	Weak contextual evidence	Investigation hint only
E2	Identity-supporting evidence	Candidate component identity
E3	Independent multi-source identity evidence	Strong identity claim
E4	Version-discriminating APK evidence	Bounded exact or version-set claim
E5	Externally validated exact evidence	Exact claim in the validation context

4.5 Evidence Quality Dimensions

4.6 Evidence Fusion, Conflict, and Abstention

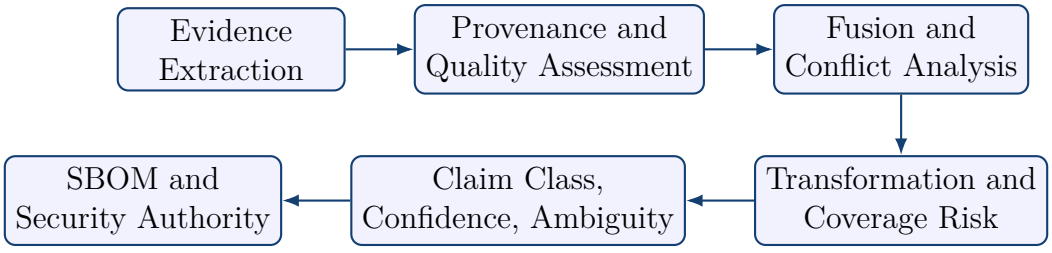


Figure 4.1: Conceptual flow of the proposed evidence-based trust framework.

4.7 Claim Classes

4.8 Validation and Evidence Lanes

4.9 Vulnerability-Claim Policy

4.10 Recovered Claim Lifecycle

4.11 PFE System as a Case Study

Selected PFE observations will be used to demonstrate controlled exact recovery, clean and partial real-APK evidence, version-band ambiguity, reference coverage versus matcher recall, source-known upper-bound analysis, failed promotions, report-only AI, and native or hidden-code boundaries.

4.12 Framework Evaluation

4.13 Limitations and Research Agenda

4.14 Conclusion

Part V

General Conclusion

Chapter 5

General Conclusion and Future Directions

5.1 General Conclusion

5.2 Answers to the Research Questions

5.3 Main Contributions

5.4 Limitations

5.5 Future Research

5.6 Closing Remarks

Appendix A

Literature Review Protocol

This appendix records version 1.0 of the structured literature mapping protocol, frozen on 21 June 2026. It is the operational companion to the methodology summarized in chapter 2. The review follows the planning, selection, quality-assessment, extraction, and synthesis principles of software-engineering review guidance [5], while explicitly acknowledging that the seed corpus was assembled iteratively before the protocol was frozen.

A.1 Research Questions

RQ-R1 Which evidence sources are used to identify Android third-party components from compiled applications?

RQ-R2 Which techniques support library identity, exact-version attribution, or both?

RQ-R3 Which compilation, optimization, obfuscation, packaging, and dependency transformations threaten each evidence source?

RQ-R4 How do studies construct ground truth and evaluate library-level, version-level, and vulnerability-related claims?

RQ-R5 How do studies represent confidence, ambiguity, provenance, abstention, partial inclusion, and conflicting evidence?

RQ-R6 Which research gaps prevent recovered component claims from becoming trustworthy and auditable SBOM statements?

A.2 Search Strategy

A.2.1 Sources and Scope

The primary search sources are ACM Digital Library, IEEE Xplore, SpringerLink, ScienceDirect, USENIX, and arXiv. Google Scholar is used for discovery and backward or forward citation chaining rather than as the sole bibliographic authority. The publication window begins in 2015 and ends on the date recorded for each search refresh. English full text is preferred; English title and abstract are mandatory for screening.

A.2.2 Concept Blocks

Table A.1: Search concept blocks

Block	Terms
Platform	Android, APK, DEX, Dalvik
Component	third-party library, TPL, software component, dependency, SBOM, software bill of materials
Task	detect*, identif*, recogn*, recover*, attribut*, fingerprint*, match*
Threat or specificity	obfuscat*, R8, ProGuard, optim*, shrink*, inlin*, version*, vulnerab*

The canonical query is:

```
(Android OR APK OR DEX OR Dalvik)
AND ("third-party library" OR TPL OR "software component" OR dependency
   OR SBOM OR "software bill of materials")
AND (detect* OR identif* OR recogn* OR recover* OR attribut*
   OR fingerprint* OR match*)
AND (obfuscat* OR R8 OR ProGuard OR optim* OR shrink* OR inlin*
   OR version* OR vulnerab*)
```

Database-specific syntax may change, but a refresh must preserve the four concept blocks and record the exact executed query, date, raw result count, deduplicated count, and selection decisions.

A.3 Inclusion and Exclusion Criteria

A.3.1 Inclusion

A publication is included when it:

1. addresses component, library, package, or version recovery from Android binaries or evaluates tools that do;
2. studies evidence obtainable from compiled artifacts, reference libraries, package metadata, or related dependency knowledge;
3. contributes a method, benchmark, empirical assessment, taxonomy, or SBOM-oriented framework relevant to compiled-artifact attribution;
4. permits extraction of at least the task, evidence source, evaluation basis, and limitations; and
5. has an accessible English title and abstract.

A.3.2 Exclusion

A publication is excluded when it:

1. relies only on source dependency manifests and does not study binary recovery;
2. concerns malware classification, clone detection, or vulnerability scanning without a component-attribution contribution;
3. is a tutorial, slide deck, blog post, product page, or duplicate;
4. reports no separable Android result;
5. lacks enough information to identify its method or evaluation basis; or
6. is a shorter version of an included extended study and contributes no distinct method, dataset, or result.

A.3.3 Screening and Evidence Tiers

Screening proceeds through deduplication, title-and-abstract screening, full-text screening, and tier assignment. The evidence tiers are:

Tier A — full-text verified A local full text has been inspected and the publication may support structured and quantitative extraction.

Tier B — supplementary record Metadata or a structured note exists, but the local full text has not yet been verified. The record may guide search and provisional gap mapping, but not precise quantitative claims.

Tier C — contextual authority Standards, official documentation, and project artifacts used for definitions or framework design. They are not counted as primary studies.

Excluded The reason for exclusion is recorded and the item is not used in synthesis.

A.4 Data Extraction Schema

Table A.2: Structured extraction fields

Dimension	Extracted information
Bibliographic identity	Title, authors, year, venue, DOI or stable URL, publication type, study family, evidence tier.
Objective and task	Research objective; family, library, class, version, SBOM, or vulnerability granularity.
Inputs and assumptions	Required APK artifacts, reference corpus, metadata, training data, and coverage assumptions.
Segmentation	Package boundaries, class hierarchy, dependency graph, clustering, or other module-decoupling method.

Dimension	Extracted information
Evidence	Names, APIs, opcodes, control flow, call/dependency graphs, literals, resources, native code, metadata, learned representations, and cross-component relations.
Decision method	Hashing, similarity, graph matching, ranking, classification, probability, learned model, or evidence fusion.
Uncertainty behavior	Threshold, confidence, competing versions, abstention, partial matching, provenance, and conflict handling.
Transformations	Renaming, package flattening, shrinking, optimization, control-flow rewriting, method inlining, repackaging, multidex, partial inclusion, native code, and dynamic loading.
Evaluation	Dataset, ground truth, evaluation unit, metrics, baselines, obfuscation setting, and independent benchmark results.
Operational support	Artifact availability, runtime, scalability, SBOM format, vulnerability mapping, and transitive-dependency handling.
Limitations and relevance	Threats to validity, negative findings, and adoption, adaptation, or rejection relevance to the framework.

Unreported information is recorded as `NR`; it is not inferred from neighboring fields or from another publication in the study family.

A.5 Quality Assessment

Tier A publications receive a score of 1, 0.5, or 0 for each question:

1. Is the research objective clear?
2. Is the method described well enough to understand or reproduce?
3. Are the dataset and ground truth described?
4. Are the task unit and metrics explicit?
5. Are transformation or obfuscation conditions stated?
6. Are identity and exact-version claims separated?
7. Is an implementation, dataset, or artifact available?
8. Are limitations or threats to validity acknowledged?

The score controls synthesis weight rather than automatic inclusion. Historically important studies may remain in the map with an explicit reporting-quality limitation.

A.6 Synthesis Rules

- Narrative and taxonomy-based synthesis is the default.
- Incompatible datasets and metrics are not pooled into a common leaderboard.

- Library-level, class-level, and version-level results remain separate.
- Original-paper results remain separate from independent benchmark results.
- Obfuscation and optimization conditions must accompany performance claims.
- Artifact availability is assessed independently of reported accuracy.
- Tier B records cannot independently support precise numerical claims.
- Identity authority, version authority, and vulnerability authority are separate.
- Explicit ambiguity or abstention is preferred to unsupported exactness.

A.7 Selected Studies

Table A.3: Frozen seed corpus and current evidence status

ID	Publication or study label	Year	Status
S01	LibScout [8]	2016	Tier A
S02	LibD [11]	2017	Tier A
S03	LibPecker [12]	2018	Tier A
S04	ORLIS [9]	2018	Tier A
S05	LibID [10]	2019	Tier A
S06	Automated Third-Party Library Detection: Are We There Yet? [6]	2020	Tier A
S07	Systematic Assessment of Android TPL Detection Tools [7]	2021	Tier A
S08	ATVHUNTER [13]	2021	Tier A
S09	LibScan [14]	2023	Tier A
S10	Empirical Study of TPL Detection in iOS and Android [18]	2024	Tier A
S11	LibHunter / Code-Optimization Impact Study [15]	2024	Tier A
S12	LibAttention [17]	2025	Tier A
S13	SAD [16]	2025	Tier A
S14	LibRadar	2016	Tier B
S15	Libra	2023	Tier B
S16	LibMD	2025	Tier B
S17	TPL-Benchmark / Revisiting TPL Detection	2025	Tier B
S18	UniBOM	2025	Tier B
S19	LibScope / Towards Global Matches	2026	Tier B
S20	ObfDetector / Advanced-Obfuscation Robustness Study	2026	Tier B

The seed corpus contains 20 publication units: 13 Tier A full texts and seven Tier B supplementary records. This count is a protocol checkpoint, not a claim that all 20 have equal evidential authority. A Tier B record is promoted only after full-text verification, bibliographic checking, structured extraction, and quality assessment.

A.8 Audit and Change Control

The current extraction records are maintained under `papers/data/`. Each search refresh must append the date, sources, query variants, raw and deduplicated result counts, inclusion and exclusion decisions, tier changes, and taxonomy effects to the master-thesis session log. Eligibility, evidence tiers, and synthesis rules may be amended only through a dated protocol change note; silent changes are prohibited.

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